A Quantitative Convergence Law for the Connes–Consani–Moscovici Zeta Spectral Triple

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Abstract

We give a quantitative convergence law for the first Riemann zero produced by the Connes–Consani–Moscovici (CCM) zeta spectral triple: a three-budget minimum in digit space, equivalently a single two-channel spectral edge, with every component carried at an explicit rigor level. In detail, the CCM construction [1] and its Connes–van Suijlekom refinement [2] produce finite real symmetric matrices whose ground state yields an approximation ν_1 to the first nontrivial Riemann zero γ_1 that sharpens as three parameters grow: a prime cutoff λ^2, a basis size N, and the working precision P. The rate of this convergence in the basis size is named the central open problem in [1]. We characterize the convergence quantitatively, in two equivalent forms. Intrinsically, the construction's relative error $|\nu_1 - \gamma_1|/|\gamma_1|$ is controlled by a single two-channel decomposition,

$$E(N, \lambda^2) \leq \varepsilon_{\text{modes}}(N, \lambda^2) + \varepsilon_{\text{primes}}(\lambda^2),$$

with both channels *defined* from the construction (the finite-N displacement at fixed cutoff and the cutoff displacement at infinite basis size), a mode channel vanishing as $N \to \infty$ and a prime channel vanishing as $\lambda^2 \to \infty$, so that $\nu_1 \to \gamma_1$ is exactly $E \to 0$ and forces both limits. The two channels are the finite-N transition and the extreme tail of a single spectral edge (the lower edge of the equilibrium measure of the archimedean field, whose bulk density is the Riemann–von Mangoldt law), so D_{nModes} and D_{Primes} are one edge read at two scales. Observed at finite working precision P, the same law is the matching-digit count $D(\lambda^2, N, P) = -\log_{10}(|\nu_1 - \gamma_1|/|\gamma_1|)$,

$$D(\lambda^2, N, P) = \min\bigl(D_{\text{Wprec}}(P),\, D_{\text{nModes}}(N, \lambda^2),\, D_{\text{Primes}}(\lambda^2)\bigr),$$

the minimum of three budgets (precision, modes, and prime content), of which precision is the instrument resolution, absent from the intrinsic form. In digit space the minimum is forced, not fitted (D is a negative logarithm of a sum of independent errors, so the largest binds), which is why earlier single-expression digit fits fail. We prove, under an explicit finite-cutoff positivity hypothesis (unconditional in the Yoshida–Bombieri small-support range), that convergence to γ_1 requires all three budgets to diverge: at a fixed cutoff the accuracy is capped at the prime ceiling $D_{\text{Primes}}(\lambda^2)$, so no basis size or precision reaches the zero and the cutoff must grow. This reduces convergence to the growth of the prime ceiling, whose leading coefficient is derived, parameter-free, from the prolate spheroidal eigenvalue asymptotic: $D_{\text{Primes}} \sim (4\pi/\ln 10)\lambda^2$, rigorously anchored from one side by a variational floor bound; the reverse inequality, that the ceiling cannot be beaten at this rate, remains open. The mode budget, the basis-size convergence rate, is the logarithm of a stretched-exponential convergence $\exp(-aN^q)$ whose exponent $q(\lambda^2)$ is the analyticity class of the eigenvector, derived from the archimedean Gamma factor with a prime correction from Mertens' theorem; a by-product is that the eigenvalue-space Galerkin exponent measured by Groskin [7] is a finite-window slice of the same curve. We state each component with an explicit rigor level, separating what is proved versus what is derived and empirically validated up to $\lambda^2 = 200$; the structural inputs are independent of [2] (Appendix A). The results have been communicated to co-authors of [1].

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1 Introduction

The Connes–Consani–Moscovici (CCM) construction [1], building on the Connes–van Suijlekom truncated Weil form [2], produces a family of finite self-adjoint operators $D_{\log}^{(\lambda,N)}$ whose spectra approximate the imaginary parts γ_k of the nontrivial Riemann zeta zeros to remarkable numerical accuracy: 55 digits at the headline configuration of [1], which we extended to 999 digits in an independent reproduction [31]. The construction has two integer-like control parameters, a prime cutoff λ^2 (the primes $p \leq \lambda^2$ enter the Weil form) and a basis size N (the matrix is $(2N+1) \times (2N+1)$), together with the working precision P in decimal digits. The rate at which the eigenvalue converges in the basis size N is named the central open problem in [1]; Groskin [7] measured a Galerkin convergence exponent empirically but disclaims deriving it, and Śliwiński [8] proves an inverse-logarithmic lower bound on the mean spectral error. We previously tested two proposed rate laws against high-precision data and found them unsupported [32].

The author's starting point is a belief the Hilbert–Pólya conjecture already expresses: that apparent randomness in the zeros is structure not yet written down. The CCM construction gives that belief a computable form. What remained open is the rate at which the computation converges, and characterizing that rate explicitly is the subject of this paper.

Definition 1. $D(\lambda^2, N, P) := -\log_{10}\big(|\nu_1 - \gamma_1|/|\gamma_1|\big)$ *is the* matching-digit count*: the number of correct leading decimal digits of the first Riemann zero* γ_1 *reproduced by* ν_1*, the first ordinate produced by the CCM construction, computed at working precision* P *via MPFR/GMP. The reference* γ_1 *is computed to* 2500 *significant digits by rigorous interval arithmetic (Arb [33]), far exceeding the largest match reported here (*≈ 1070 *digits at* $\lambda^2 = 200$*), so the reported counts are genuine and not reference-limited; its leading digits agree with the standard tabulations of [34].*

Notation (each object is defined where cited; collected here for reference)	
$E = \lvert\nu_1 - \gamma_1\rvert/\lvert\gamma_1\rvert$	relative error of the first ordinate; $D = -\log_{10} E$ (Def. 1)
$\varepsilon_{\text{modes}},\ \varepsilon_{\text{primes}}$	intrinsic error channels, eq. (3)
$D_{\text{Wprec}},\ D_{\text{nModes}},\ D_{\text{Primes}}$	per-source digit budgets (Sections 4–6)
$\varepsilon^\infty(\lambda)$	smallest eigenvalue of the continuum form at cutoff λ
$D_{\text{eig}} = -\log_{10} \varepsilon^\infty$	eigenvalue floor (Section 6)
$D_{\text{edge}} = D_{\text{Primes}} \cdot g$	the single edge function (Section 3.1)

The same number is read as a *ceiling* or a *floor* depending on the direction of view. D_{Primes} is a ceiling on the achievable digit count and, equally, the depth of the accuracy floor at the cutoff; both usages refer to the one object.

Our central result, the *Andrews CCM Convergence Law*, has two equivalent forms. Its intrinsic (spectral) form is the two-channel decomposition

$$\boxed{E(N, \lambda^2) \leq \varepsilon_{\text{modes}}(N, \lambda^2) + \varepsilon_{\text{primes}}(\lambda^2)} \tag{1}$$

with the channels defined in Section 3, (3) (an equality up to sign cancellation, confined to crossover seams; Prop. 1). The two channels are the finite-N transition and the extreme tail of a single prolate-type spectral edge (Section 3.1); convergence $\nu_1 \to \gamma_1$ is exactly $E \to 0$. Its finite-precision measurement is the matching-digit count

$$\boxed{D(\lambda^2, N, P) = \min\big(D_{\text{Wprec}}(P),\ D_{\text{nModes}}(N, \lambda^2),\ D_{\text{Primes}}(\lambda^2)\big)} \tag{2}$$

in which D_{Wprec} is the instrument resolution alone. Since D is the relative error on a logarithmic scale, the law is a quantitative convergence statement: $D \to \infty$ is precisely $\nu_1 \to \gamma_1$, and the three budgets are the independent mechanisms that drive (or stall) that convergence (Prop. 4). Five points distinguish it from prior empirical fits:

1. **The minimum is forced, not fitted** (Section 3). In digit space D is a negative logarithm of an error bounded by a sum of operationally independent channels, so the largest binds and D is at least the minimum of the three per-source budgets, unconditionally, and equals it up to an $O(1)$ term absent cancellation (Prop. 1). This is the structural reason earlier single-formula digit fits failed. The precision budget is measurement, not mathematics: removing it leaves the single intrinsic edge (1), whose two channels (modes, primes) are the only mathematical content.

2. **Each budget is analyzed from the construction**, not guessed. The precision budget follows from elementary round-off analysis; the prime ceiling from the prolate eigenvalue asymptotic; and the mode budget from the analyticity of the Weil-form eigenvector.

3. **Every claim carries an explicit rigor level** (Table 2), separating the rigorously proved parts from those derived and validated against high-precision computation. We do not label the whole law a theorem; we state precisely which pieces are.

4. **It yields a convergence criterion** (Prop. 4): the prime cutoff is an unbeatable ceiling. At a fixed λ^2 no basis size or precision reaches the zero, so convergence to γ_1 forces $\lambda^2 \to \infty$, which turns the qualitative convergence of [1] into an explicit, conditional rate.

5. **No prior unified law, to our knowledge.** Earlier attempts are single closed-form fits of the digit count, and they fail: we tested two proposed rate laws against high-precision data and found them unsupported [32]. Groskin [7] likewise reports that no single smooth-convergence law fits his data and declines to assert a rate, and Śliwiński [8] establishes only a lower bound on the mean error. The unified characterization given here, the intrinsic two-channel edge (1) and its finite-precision measurement (2), is, to the best of the author's knowledge, the first working closed-form convergence law for the construction. Its unity is the point: one spectral edge (Section 7) underlies both the mode ramp and the prime ceiling, which is why a single expression can hold where earlier per-fits could not.

The deepest new content is the mode budget D_{nModes} (Section 5): we show the matching-digit ramp is the logarithm of a stretched-exponential convergence, identify its exponent with the analyticity class of the eigenvector, and derive that exponent end to end. A by-product is that the eigenvalue-space Galerkin exponent measured by Groskin [7] is a finite-window slice of the same curve. That the mode ramp and the prime ceiling are not two mechanisms but one spectral edge at two scales, the lower edge of an equilibrium measure whose bulk is the Riemann–von Mangoldt density, is the subject of Section 7.

2 The construction and the object D

We recall the construction of [1, 2]. Let $\lambda > 1$ and $L = 2\ln\lambda = \ln\lambda^2$. On $\mathcal{H}_\lambda = L^2([\lambda^{-1}, \lambda], du/u)$ the functions $V_n(u) = L^{-1/2}e^{2\pi i n \ln(\lambda u)/L}$ form an orthonormal basis. The Weil quadratic form QW_λ, restricted to $\mathrm{span}\{V_n : |n| \le N\}$, is a real symmetric $(2N{+}1) \times (2N{+}1)$ matrix whose archimedean part carries the local factor at infinity and whose arithmetic part is a finite sum over prime powers $p^j \le \lambda^2$. Its smallest eigenvalue ε_N has an even eigenvector ξ, and the positive zeros of the associated Fourier–Mellin function $\widehat{\xi}$ approximate the ordinates γ_k [1].

Two facts about $\widehat{\xi}$ are used below: $\widehat{\xi}$ is entire, and (in the critical case) all its zeros are real and coincide with the spectrum. Both are proven *self-contained* in [1]: the closed form of the transform is [1, Prop. 5.9] (rederived from scratch in Appendix A), and entirety/reality is [1, Thm. 5.10], whose proof chain (Lemmas 5.2 and 5.4 there: the matrix structure $\tau_{ij} = (b_i - b_j)/(i-j)$, an elementary commutator identity, and self-adjointness in the induced inner product) is complete within [1]. The Carathéodory–Fejér–type reality mechanism *originates* in the Connes–van Suijlekom paper [2], which we cite as the historical source; no statement used in this paper depends on it (Appendix A). An independent classical foundation for the truncated Weil form predates both: Weil's positivity criterion, Yoshida's unconditional positivity theorem for test functions of sufficiently small support [9], Bombieri's study of the truncated quadratic form and the eigenvalues of its finite sections [10], and Suzuki's screw-function framework [13, 14, 5], which unifies all of these by means of continuous functions.

The eigenvector ξ is even and, in the regime we study, the smallest eigenvalue is simple, the even-symmetry verified numerically across 38 configurations spanning $\lambda^2 = 13$–1000 in [31].

The object of study, $D(\lambda^2, N, P)$, is a digit count: $D = -\log_{10}(\text{relative error})$. The relative error itself, $E = |\nu_1 - \gamma_1|/|\gamma_1|$, is the intrinsic object of the law (1); D is its measurement in digits, and the arithmetic of that measurement is what makes the minimum structure of (2) inevitable.

3 The three-budget skeleton

The computed approximation ν_1 carries error from three sources: finite working precision (round-off), finite basis size (mode truncation), and finite prime cutoff (the intrinsic ceiling at $N, P \to \infty$). We now *define* the channels, rather than posit their additivity. Write $\nu_1(N, \lambda^2)$ for the first ordinate of the exact (infinite-precision) finite-N operator and $\nu_1(\infty, \lambda^2)$ for its $N \to \infty$ limit at fixed cutoff, and set

$$\varepsilon_{\text{modes}}(N, \lambda^2) := \frac{|\nu_1(N, \lambda^2) - \nu_1(\infty, \lambda^2)|}{|\gamma_1|}, \qquad \varepsilon_{\text{primes}}(\lambda^2) := \frac{|\nu_1(\infty, \lambda^2) - \gamma_1|}{|\gamma_1|}, \tag{3}$$

the finite-N displacement at fixed cutoff and the cutoff displacement at infinite basis size, with $\varepsilon_{\text{round}}(P)$ the round-off perturbation of the computed ν_1 relative to the exact finite-N value. The triangle inequality then gives, for the computed relative error,

$$E \;\le\; \varepsilon_{\text{round}}(P) + \varepsilon_{\text{modes}}(N, \lambda^2) + \varepsilon_{\text{primes}}(\lambda^2),$$

an inequality, not an identity: the signed displacements can in principle cancel. The channels are *independent* in a precise operational (not probabilistic or structural) sense: each answers to its own parameter (round-off to P, truncation to N, prime content to λ^2), so any one can be driven below the others at will. (Structurally they are coupled: $\varepsilon_{\text{modes}}$ depends on λ^2 through both the exponent q and the Shannon coordinate of Section 5, and below the precision floor round-off corrupts the eigenvector nonlinearly, Section 4.1; the operational sense is all Prop. 1 uses.)

Proposition 1 (The minimum is forced)**.** *Let $\varepsilon_1, \varepsilon_2, \varepsilon_3 \ge 0$ and let $E \ge 0$ satisfy $E \le \Sigma := \sum_i \varepsilon_i$. Then, unconditionally,*

$$D := -\log_{10} E \;\ge\; -\log_{10}\Sigma \;\ge\; \min_i\bigl(-\log_{10}\varepsilon_i\bigr) - \log_{10} 3,$$

since $\Sigma \le 3\max_i \varepsilon_i$. If in addition no cancellation occurs — quantitatively, $E \ge \theta \max_i \varepsilon_i$ for some fixed $\theta \in (0, 1]$ — then also $D \le \min_i(-\log_{10}\varepsilon_i) + \log_{10}(1/\theta)$, whence $D = \min(D_{\text{Wprec}}, D_{\text{nModes}}, D_{\text{Primes}}) + O(1)$.

The lower bound, that the computed digit count is at least the binding budget minus half a digit, is a rigorous consequence of the triangle inequality alone. The matching upper bound holds *absent cancellation* between the signed displacements. Such cancellation is confined to the narrow crossover seams where two budgets are comparable (away from a seam a single channel dominates E outright and θ is near 1), and no seam in the 5,000-plus experiments of this program shows D exceeding the binding minimum by more than the stated $O(1)$; with that caveat recorded once, we write the law as an equality throughout. The $O(1)$ term is at most $\log_{10} 3 < 0.48$ of a digit and lives only at the seams, so the law's content is that the *binding mechanism* changes across the parameter space (round-off, mode truncation, then prime content), not that D is literally non-smooth (it is smooth: a negative logarithm of a near-sum).

This is why, in digit space, D is a minimum rather than a sum or product, and why global single-expression fits of the digit count fail in practice: a single elementary term cannot simultaneously hold a hard precision wall, a prime ceiling, and a λ^2-coupled mode ramp. In the intrinsic form the error is, by contrast, the two-channel decomposition (1); the digit-space minimum is what the negative logarithm of that near-sum becomes, the largest term binding. The junctions differ in character. The precision junction is a hard wall, since round-off is a clamp, so its corner is sharp. The mode→prime junction, by contrast, is a wide, smooth saturation (the tanh of Section 5): D_{nModes} is the mode-limited digit count, which rises along its pre-saturation ramp and is capped by the ceiling, $D_{\text{nModes}} \le D_{\text{Primes}}$, with equality as $N \to \infty$.

The starvation protocol. Each budget is isolated experimentally by *starvation*: to read D_i directly, set the other two parameters large enough that their budgets exceed D_i, so the channel of interest dominates E outright and the cancellation caveat of Prop. 1 is vacuous. This protocol is the experimental mirror of the minimum structure. It is what makes the digit form falsifiable rather than descriptive, since each budget's functional form is measured in a regime where it alone binds. We use it throughout Sections 4–6.

3.1 The two forms of the law

Of the three budgets, only two (mode truncation and prime cutoff) are properties of the finite operator; round-off is a property of the computation. Separating them gives the two forms stated in Section 1.

Intrinsic (spectral) form. The mathematical content is the two-channel decomposition (1), with $\varepsilon_{\text{modes}} \sim \exp(-aN^q) \to 0$ as $N \to \infty$ (Section 5) and $\varepsilon_{\text{primes}} \to 0$ as $\lambda^2 \to \infty$ (Section 6). It makes no reference to precision, and $\nu_1 \to \gamma_1$ is exactly $E \to 0$. Vanishing of both channels forces $E \to 0$ unconditionally; conversely, sustained convergence $E \to 0$ forces both channels to vanish, since a bounded cutoff pins $\varepsilon_{\text{primes}}$ away from zero (Prop. 4) and the fixed-λ^2 mode displacement, vanishing as $N \to \infty$, cannot cancel it in the limit — accidental cancellation can occur only at isolated parameter values, not along a convergent path.

Observed (digit) form. A computation at working precision P resolves E only down to its round-off floor, so the measured matching-digit count is

$$D(\lambda^2, N, P) = \min\bigl(D_{\text{Wprec}}(P), -\log_{10} E(N, \lambda^2)\bigr) = \min\bigl(D_{\text{Wprec}}, D_{\text{nModes}}, D_{\text{Primes}}\bigr) + O(1), \quad (4)$$

the second equality by Prop. 1. Here $D_{\text{Wprec}}(P)$ is the instrument resolution: it is no property of ν_1, γ_1, or the operator, and as $P \to \infty$ it disappears, returning the intrinsic form in digit units. Thus the boxed law (2) is the finite-precision *measurement* of the intrinsic law (1).

One edge, not two budgets. The mode and prime channels are the two ends of a single spectral edge. Dividing out the ceiling (Section 5) writes $D_{\text{nModes}} = D_{\text{Primes}} \cdot g$ with $g = \tanh\bigl((c\,N/(\lambda^2 \ln \lambda^2))^q\bigr) \uparrow 1$ as $N \to \infty$, so $D_{\text{Primes}} = \lim_{N\to\infty} D_{\text{nModes}}$ is the $N \to \infty$ ceiling of the mode ramp, not a separate term; equivalently the digit law is a single edge function $D_{\text{edge}}(N, \lambda^2) = D_{\text{Primes}} \cdot g$ capped by precision, $D = \min(D_{\text{Wprec}}, D_{\text{edge}}) + O(1)$. Correspondingly the intrinsic error (1) is one edge: $\varepsilon_{\text{modes}}$ is its finite-N transition (the Landau–Widom plunge, Section 5) and $\varepsilon_{\text{primes}}$ is its extreme tail (the prolate asymptotic, Section 6). That these are literally the transition and the tail of one band edge, the lower edge of the archimedean equilibrium measure, is established in Section 7.

4 The precision budget D_{Wprec}

Finite working precision P (decimal digits) introduces relative round-off $\sim 10^{-P}$, amplified by the eigenproblem's condition number κ (set by the gap of ε_N to the rest of the spectrum and independent of P). Hence

$$D_{\text{Wprec}}(P) = P - \log_{10}\kappa = P + O(1), \tag{5}$$

with leading coefficient exactly 1. The $O(1)$ term is implementation dependent: the toolkit used here allocates $\lceil P \cdot \ln 10/\ln 2 \rceil + 16$ working bits, the 16 guard bits contributing a small positive offset. Isolating this budget (large λ^2 and N) and sweeping $P \in \{100, 200, 300, 400, 700, 1000\}$ gives a measured slope $\partial D/\partial P = 1.000 \pm 0.001$. The exact offset is not a property of the construction and we do not assign it physical meaning. *Status: T1; the coefficient* 1 *follows from elementary numerical analysis.*

4.1 Symmetry, positivity, and the precision floor

A question the precision floor settles, and one easy to answer wrongly without it, is whether the even symmetry of the ground eigenvector, and the positivity of ε_N, are genuine features of the operator or merely of the regime above the floor. The CCM construction [1] posits a simple smallest eigenvalue with an *even* eigenvector (its Step 1) and enforces this with a forced-even projection. The reproduction of [31] tested that projection directly: across 38 configurations ($\lambda^2 = 13$–1000, $N = 10$–800, spanning both sides of the precision floor) with the projection *disabled*, the natural smallest eigenvector converged to the very vector the projection would have produced, to *bit-identical* eigenvalues, at every configuration above the floor. The forced-even projection is thus empirically a no-op above the floor: even symmetry is a property the operator exhibits unbidden, not an artifact of the projection or of the above-floor regime.

The qualifier "above the floor" is essential, and it is precisely where the budget of this section bites. When the target eigenvalue lies below D_{Wprec}, smaller than the round-off that the working precision can resolve, the computed eigenvector is dominated by numerical noise: it can present spurious mixed symmetry, and the smallest computed eigenvalue can carry a spurious sign. These are artifacts of precision starvation, not properties of the operator; raising P past the floor dissolves them, and the natural eigenvector is again even, into the deep near-zero spectrum. (This is also the regime in which the round-off channel of Section 3 stops acting as an additive perturbation of ν_1: below the floor it corrupts the eigenvector nonlinearly, so the decomposition (3), like the symmetry and positivity statements of this subsection, is an above-floor statement.)

This even, positive ground state was documented in our reproduction [31]. In subsequent correspondence, and in a revision of his own paper [7], Groskin independently corroborated it by a different route: the sign changes his separate quadrature implementation had shown among the deepest eigenvalues at a finite archimedean cutoff T were identified as a finite-T artifact that vanishes as T is increased, consistent with the naturally even, positive ground state of [31]. Working precision P and archimedean cutoff T are therefore independent controls: a deep-spectrum value is trustworthy only when it is stable across two settings of *each*, not when it merely holds steady in one. The even, positive ground state persists into the deep near-zero spectrum; apparent breakdowns are finite-precision or finite-cutoff artifacts, not features of the construction.

5 The mode budget D_{nModes}

This is the convergence rate in N, the open problem of [1]. In the intrinsic form it is the mode channel $\varepsilon_{\text{modes}}$; as a digit count it is the rising transition of the equilibrium-measure edge

(Section 7), climbing toward the prime ceiling. Dividing out that ceiling (Section 6), write the ramp ratio $g = D_{\text{nModes}}/D_{\text{Primes}} \in [0, 1]$. We claim

$$D_{\text{nModes}}(N, \lambda^2) = D_{\text{Primes}}(\lambda^2) \cdot \tanh\Big(\big(c\,N/(\lambda^2 \ln \lambda^2)\big)^{q(\lambda^2)}\Big), \qquad q(\lambda^2) = \frac{2}{2\gamma + \ln\ln\lambda^2}. \tag{6}$$

We build this in four steps: the stretched-exponential meaning of the ramp (5.1), the archimedean leading rate (5.2), the prime correction giving q (5.3), and the Shannon coordinate (5.4).

5.1 The ramp is a stretched exponential

Proposition 2 (Stretched-exponential convergence and rate unification)**.** *Suppose the pre-saturation mode budget satisfies $D_{\text{nModes}} = (a/\ln 10)\, N^q(1 + o(1))$ for some $a = a(\lambda^2) > 0$ and exponent $q = q(\lambda^2)$, and that the Weil-form ground eigenvalue is simple with even eigenvector. Then:*

- *(i)* (Stretched-exponential.) *The eigenvalue/zero error obeys $|\nu_1 - \gamma_1|/|\gamma_1| = \exp(-aN^q)(1 + o(1))$; equivalently q is the analyticity class of the eigenvector ($q = 1$ exponential / strip-analytic, $0 < q < 1$ sub-exponential but faster than any polynomial, $q \to 0$ finite Sobolev) — on the classical scale, $q = 1/\mu$ with μ the Gevrey index of the eigenvector [23].*
- *(ii)* (Rate unification.) *Any local power-law fit* err $\sim C\,N^{-2s}$ *over a window about N returns the* growing *local exponent $2s_{\text{loc}}(N) = -\,d\ln\text{err}/d\ln N = a\,q\,N^q$. Hence the eigenvalue-space Galerkin exponent $s(c)$ of [7] is the local logarithmic slope of this same stretched-exponential — not a fixed Sobolev index — and the digit-space ramp and the eigenvalue-space rate are one object in two coordinates. The factor 2 is the eigenvalue-approximation identity $|\varepsilon - \varepsilon_N| = O(\|\xi - \xi_N\|^2)$ for a self-adjoint operator with simple ground state [30].*

Proof. (i) $D = -\log_{10}(\text{err}) \Rightarrow \text{err} = 10^{-D} = \exp(-aN^q)$. (ii) $2s_{\text{loc}} = -\,d\ln\text{err}/d\ln N = d(aN^q)/d\ln N = aqN^q$; the squaring is the Rayleigh-quotient second-order identity at a simple eigenvalue [30]. A degenerate ground state would give only first-order dependence and no factor 2. □

Thus q is not a fitting knob but the *analyticity class* of the eigenvector: $q = 1$ is pure exponential decay (analytic in a strip), $0 < q < 1$ is sub-exponential but faster than any polynomial, and $q \to 0$ is finite Sobolev (N^{-2s}). In standard terminology this scale is the *Gevrey scale*: the Gevrey class G^μ, $\mu \geq 1$, is characterized exactly by Fourier-coefficient decay $\exp(-c\,|k|^{1/\mu})$, reducing to strip-analytic functions at $\mu = 1$ [23], so the statement $|\xi_k| \sim \exp(-c\,k^q)$ is the statement $\xi \in G^{1/q}$ (and no better). The open "variance-to-exponent map" of Section 5.3 is thereby a precisely posed regularity lemma on a classical scale: *prove that the ground eigenvector lies in the Gevrey class $G^{1/q(\lambda^2)}$ sharply, with $1/q = \gamma + \frac{1}{2}\ln\ln\lambda^2$.* The measured exponents lie just below 1 and decrease with λ^2 (Table 1), the signature of an eigenvector that is strip-analytic in its archimedean part and roughened by the prime content.

The force of part (i) is that it converts the open problem of *measuring* a convergence rate into the analytic question of *identifying* the eigenvector's regularity: the single exponent q fixes the entire pre-saturation curve. It also explains why fixed power-law (Sobolev) fits are structurally the wrong model (they are the $q \to 0$ limit, whereas the operator sits near $q = 1$), and hence why such fits drift and resist extrapolation (Section 9).

Part (ii) identifies the Galerkin exponent $s(c)$ measured by Groskin [7] in eigenvalue space as the local logarithmic slope of the present stretched-exponential, sampled at his window; the data-level reconciliation this affords is taken up in Section 9. The squaring that links them needs the ground eigenvalue to be *simple*, which is not assumed but established as the naturally-even, isolated ground state verified across 38 configurations in the reproduction [31]; were the state degenerate, the eigenvalue error would be only first order in the eigenvector error and the factor 2, equivalently the $2s$ in Groskin's rate, would not appear. *Status: T1 (given the pre-saturation ramp form as hypothesis).*

5.2 The archimedean leading rate $\delta = \pi/4$

The first positive zero ν_1 of the Fourier–Mellin function $\widehat{\xi}(z)$ converges to γ_1 at a rate set by the analyticity strip of the *limiting* Fourier–Mellin function, governed by the archimedean factor of the Weil form. That factor is the completed real Gamma factor $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$: its logarithmic derivative $\psi(\frac{1}{4}+\frac{iz}{2}) \sim \ln(z/2)$ is precisely the $\ln|z|$ multiplier identified by Suzuki [5], and the factor itself decays as $|\Gamma(\frac{1}{4}+\frac{iz}{2})| \sim \exp(-\pi|z|/4)$ by Stirling. Hence the strip half-width is

$$\delta = \pi/4. \tag{7}$$

This value has a clean structural origin, one that also shows why the scales $\frac{1}{2}$ and $\pi/4$ both attach to the same archimedean factor. On the critical line the archimedean log-derivative has the large-z expansion

$$\psi\big(\tfrac{1}{4} + \tfrac{iz}{2}\big) = \log\frac{z}{2} + i\,\frac{\pi}{2} + o(1), \qquad z \to +\infty, \tag{8}$$

since the argument $\frac{1}{4} + \frac{iz}{2}$ runs in the imaginary (critical-line) direction, so $\arg\big(\frac{1}{4} + \frac{iz}{2}\big) \to \pi/2$. Its real and imaginary parts carry the two scales:

- the *real* part $\operatorname{Re}\psi(\frac{1}{4} + \frac{iz}{2}) \to \log(z/2)$ is the symbol, Suzuki's $\log|z|$ multiplier [5], whose analyticity strip has half-width $\frac{1}{2}$ (the digamma poles at $z = \pm i/2$);
- the *imaginary* part $\operatorname{Im}\psi(\frac{1}{4} + \frac{iz}{2}) \to \frac{\pi}{2}$ sets the envelope, through

$$\frac{d}{dz}\log\big|\Gamma_{\mathbb{R}}(\tfrac{1}{2} + iz)\big| = -\tfrac{1}{2}\operatorname{Im}\psi\big(\tfrac{1}{4} + \tfrac{iz}{2}\big) \longrightarrow -\tfrac{1}{2}\cdot\tfrac{\pi}{2} = -\tfrac{\pi}{4}.$$

The strip half-width and the envelope rate are therefore the real and imaginary parts of the *one* analytic function $\log\Gamma_{\mathbb{R}}(\frac{1}{2} + iz)$, a Hilbert (Kramers–Kronig) conjugate pair since $\Gamma_{\mathbb{R}}$ is zero-free (minimum phase): its log-magnitude and phase determine each other. In one identity,

$$\delta = \frac{\pi}{4} = \frac{1}{2}\cdot\frac{\pi}{2} = \tfrac{1}{2}\arg(i), \tag{9}$$

half the right angle of the critical-line direction. The two scales $\frac{1}{2}$ and $\pi/4$ are thus not independent constants but conjugate parts of the single factor $\Gamma_{\mathbb{R}}$: the strip half-width $\frac{1}{2}$ and its harmonic conjugate $\pi/4$.

Two points delimit what this establishes. It fixes the *value* $\delta = \pi/4$ as a property of the limiting $\Gamma_{\mathbb{R}}$ factor: δ is the half-width of the strip carrying the zeros of the limiting Fourier–Mellin function, and governs the zero convergence $\nu_1 \to \gamma_1$ (with the squaring of Prop. 2(ii)); it is not the decay rate of the finite eigenvector's Galerkin coefficients, a distinct band-limited, prime-modulated quantity (≈ 0.71 at $\lambda^2 = 100$), and the finite $\widehat{\xi}$, of exponential type $L/2$ via its $\sin(zL/2)$ factor, is not literally the completed ξ. And it is structural motivation rather than a full derivation: obtaining $\pi/4$ from the symbol alone would require its conjugate part, which appears only through the analytic completion $\Gamma_{\mathbb{R}}$, that is, through the finite ground-state transform inheriting that factor, the realization step that remains open.

Lemma 1 (Lattice sampling of the coefficients)**.** *The Fourier–Mellin transform of the N-mode eigenvector has the closed form $\widehat{\xi}_N(z) = 2L^{-1/2}\sin(zL/2)\sum_{|j|\le N}\xi_j/(z-2\pi j/L)$ ([1, Prop. 5.9]; rederived from scratch in Appendix A). On the frequency lattice $z_k = 2\pi k/L$,*

$$\widehat{\xi}_N(2\pi k/L) = (-1)^k\sqrt{L}\,\xi_k \quad (|k| \le N), \qquad \widehat{\xi}_N(2\pi k/L) = 0 \quad (|k| > N),$$

so $|\xi_k| = L^{-1/2}\,\big|\widehat{\xi}_N(2\pi k/L)\big|$, with $L = \ln\lambda^2$ the period of the basis $\{V_n\}$.

Proof. At $z_k = 2\pi k/L$ the factor $\sin(zL/2) = (-1)^k \sin\big((z - z_k)L/2\big)$ has a simple zero, so every $j \neq k$ term vanishes and the $j = k$ pole is removable: $\lim_{z\to z_k} 2L^{-1/2}(-1)^k \sin\big((z-z_k)L/2\big)/(z-z_k) = (-1)^k\sqrt{L}$. For $|k| > N$ there is no $j = k$ term and the common factor $\sin(z_k L/2) = 0$. □

By Lemma 1 the coefficients are the lattice samples of $\widehat{\xi}_N$ at spacing $2\pi/L$ with $L = \ln\lambda^2$, so the strip half-width (7) gives a coefficient tail $\exp(-\delta(2\pi/L)N)$; with the squaring of Prop. 2(ii) the leading digit slope is

$$\frac{\partial D_{\text{nModes}}}{\partial N} = \frac{2\delta\,(2\pi/L)}{\ln 10} = \frac{\pi^2}{\ln 10\ \ln\lambda^2} \approx \frac{4.286}{\ln\lambda^2}, \tag{10}$$

a parameter-free prediction. At $\lambda^2 = 7$, where few primes enter and $q \approx 1$, (10) gives 2.203 against a measured small-N slope of 2.264, agreement to 3% with no free parameters. At larger λ^2 the measured early slope runs above (10) exactly as the $q < 1$ concavity (below) requires.

What *is* rigorous is the transfer from strip convergence to zero displacement, which we now record; it confines the entire open content of $\delta = \pi/4$ to a coefficient-decay estimate.

Lemma 2 (Zero displacement from strip convergence)**.** *Let F be holomorphic on a neighborhood of the closed disc $\overline{D}(\gamma_1, r)$ with a simple zero at γ_1, no other zero in the disc, and $|F(z)| \geq \frac{1}{2}|F'(\gamma_1)|\,|z-\gamma_1|$ on the disc (true for r small enough, by continuity of F'). Let G be holomorphic on the disc with $\eta := \sup_{\overline{D}(\gamma_1,r)} |G - F|$. If $\rho := 2\eta/|F'(\gamma_1)| \leq r$, then G has exactly one zero ν in $D(\gamma_1, \rho)$, and*

$$|\nu - \gamma_1| \ \leq\ \frac{2\,\eta}{|F'(\gamma_1)|}.$$

Proof. On the circle $|z - \gamma_1| = \rho$ one has $|F(z)| \geq \frac{1}{2}|F'(\gamma_1)|\rho = \eta \geq |G(z) - F(z)|$, with strict inequality after shrinking ρ infinitesimally if equality holds; Rouché's theorem then gives G the same number of zeros as F in the disc, namely one. □

Applied with $F = \widehat{\xi}_\infty$ (the fixed-λ^2 limit transform, whose zeros are real and simple in the critical case, the Hermite–Biehler structure underlying the de Branges-space treatment of the Weil form [14, 13]) and $G = \widehat{\xi}_N$, the lemma converts uniform convergence on a neighborhood of γ_1 inside the strip into zero displacement with the explicit constant $2/|\widehat{\xi}'_\infty(\gamma_1)|$.

A second transfer can be made rigorous as well: from the coefficient tail to the transform. The subtlety is that the finite-section eigenvector is not the truncation of the limit eigenvector, so the transform difference carries a retained-coefficient error in addition to the discarded tail; both pieces are controlled by the tail alone.

Lemma 3 (Tail-to-transform transfer)**.** *Fix λ^2, assume the ground state of the continuum form QW_λ is simple with spectral gap $g(\lambda^2) > 0$, and suppose the limit eigenvector satisfies $|\xi_j^{(\infty)}| \leq C\exp(-c\,j^q)$ for some $c > 0$, $0 < q \leq 1$. Let $r > 0$ be small enough that $\overline{D}(\gamma_1, r)$ has positive distance $d(\lambda^2, r)$ from the frequency lattice $\{2\pi j/L\}$. Normalize the eigenvectors and fix phases by $\langle \xi^{(N)}, \xi^{(\infty)}\rangle \geq 0$. Then*

$$\sup_{|z-\gamma_1|\leq r} \big|\widehat{\xi}_N(z) - \widehat{\xi}_\infty(z)\big| \ \leq\ \exp\!\big(-c\,N^q(1 + o(1))\big), \qquad N \to \infty.$$

Proof. Write P_N for the projection onto $\operatorname{span}\{V_j : |j| \leq N\}$ and $Q_N = I - P_N$. With ℓ^1 coefficients the closed form of Lemma 1 extends to the limit: the series $\widehat{\xi}_\infty(z) = 2L^{-1/2}\sin(zL/2)\sum_j \xi_j/(z - 2\pi j/L)$ converges locally uniformly (lattice poles removable as before) and defines an entire function. Split $\widehat{\xi}_N - \widehat{\xi}_\infty = \widehat{T}[\xi^{(N)} - P_N\xi^{(\infty)}] - \widehat{T}[Q_N\xi^{(\infty)}]$, with $\widehat{T}$ the linear coefficients-to-function map. *Tail piece:* on the disc $|\sin(zL/2)| \leq \cosh(rL/2)$ and $|z-2\pi j/L| \geq \pi|j|/L$ for $|j|$ large, so $|\widehat{T}[Q_N\xi^{(\infty)}]| \leq C_1\sum_{|j|>N} e^{-cj^q}/|j| \leq C_2 N^{-q}e^{-cN^q}(1+o(1))$, by comparison with $\int_N^\infty e^{-ct^q}\,dt$. *Head piece:* the Galerkin residual collapses to the tail,

$$\big(QW_\lambda^N - \varepsilon^\infty\big)\big(P_N\xi^{(\infty)}\big) = P_N QW_\lambda \xi^{(\infty)} - P_N QW_\lambda Q_N\xi^{(\infty)} - \varepsilon^\infty P_N\xi^{(\infty)} = -P_N QW_\lambda Q_N\xi^{(\infty)},$$

using $QW_\lambda\xi^{(\infty)} = \varepsilon^\infty\xi^{(\infty)}$. Row bounds for the form (diagonal $O(\log j)$ from the archimedean symbol, off-diagonal rows uniformly ℓ^2, prime part bounded) give $\|QW_\lambda Q_N\xi^{(\infty)}\|_2 \leq C_3\sum_{|j|>N} e^{-cj^q}\log j$, the same exponential up to a logarithm. The section's gap persists, $g_N \geq g/2$ for N large, by standard Galerkin spectral approximation for operators with discrete spectrum [30] together with the monotone convergence of [1, Prop. 3.4]; the residual bound then yields $\|\xi^{(N)} - P_N\xi^{(\infty)}\|_2 \leq 2\|P_N QW_\lambda Q_N\xi^{(\infty)}\|_2/g_N$ up to a normalization correction of order $\|Q_N\xi^{(\infty)}\|_2^2$. Finally $|\widehat{T}[e]| \leq C_4\sqrt{2N+1}\,\|e\|_2$ on the disc, using the lattice distance $d(\lambda^2, r)$. All polynomial and logarithmic factors are absorbed into the $(1+o(1))$ in the exponent. □

Corollary 1. *Under the hypotheses of Lemma 3, $|\nu_1(N,\lambda^2)-\nu_1(\infty,\lambda^2)| \leq \exp(-c\,N^q(1+o(1)))$: the coefficient tail alone forces the zero displacement at the single exponent, by Lemma 3 followed by Lemma 2.*

The open content of the archimedean rate is thereby confined to the following statement, recorded in three clauses that separate the objects involved: the eigenvector tail of the full form, the archimedean identification of its rate, and the doubling beyond Corollary 1.

Conjecture 1 (Coefficient decay and zero displacement)**.** *Fix λ^2, let $q = q(\lambda^2)$ be the exponent of Section 5.3, and let $\xi^{(\infty)}$ denote the limit eigenvector at cutoff λ^2.*

(i) (Gevrey tail, full form.) *There exist $c(\lambda^2) > 0$ and C with*

$$|\xi_k^{(\infty)}| \;\leq\; C\,\exp\bigl(-c(\lambda^2)\,k^q\bigr).$$

(ii) (Leading rate.) *$c(\lambda^2)\,L/(2\pi) \to \delta = \pi/4$ as $\lambda^2 \to \infty$, with δ the $\Gamma_\mathbb{R}$ envelope rate of Section 5.2. The approach is from below: at finite cutoff the small-k rate sits under δ and the deficit shrinks as the cutoff grows.*

(iii) (Doubling at the zero.) *The first-zero displacement obeys*

$$|\nu_1(N,\lambda^2) - \nu_1(\infty,\lambda^2)| \;\leq\; C\,\exp\bigl(-2\,c(\lambda^2)\,N^q(1+o(1))\bigr),$$

twice the tail exponent of (i).

Three remarks delimit the clauses. Clause (i) is the Gevrey membership of Section 5.3 read per coefficient. Clauses (ii) and (iii) together recover the parameter-free slope (10) at $q \approx 1$, the prediction validated to 3% at $\lambda^2 = 7$; the finite-λ^2 coefficient rate (≈ 0.71 at $\lambda^2 = 100$, against $\delta \approx 0.785$) is then a data point for the finite-cutoff deficit that clause (ii) asserts, not a discrepancy. The author's coefficient-level measurements across $\lambda^2 = 13$–200 show the small-k rate climbing toward $\pi/4$ as the cutoff grows, with the mid-band rate below it. What controls the full profile is open and under study. Clause (iii) carries the doubling, and its open content is now exactly the factor 2: by Lemma 3 and Corollary 1, clause (i) already implies the displacement bound at the single exponent $c(\lambda^2)N^q$, rigorously, while the observed displacement decays at twice it. The mechanism, whether first-order smallness of the transform difference at the zero itself or the second-order eigenvalue identity of Prop. 2(ii) transported through the spectrum–zeros identification, is part of what must be proven. None of the clauses is a citation target: no classical result supplies the $\Gamma_\mathbb{R}$ envelope for the finite ground-state transform (the realization step, in which the finite transform inherits the completed factor), so this is new analysis, precisely posed. *Status: T2 modulo Conjecture 1, the tier carried by the displacement transfer Lemma 2 (T1); the open item is typed new analysis in Table 2, with clause (ii) the specific home of $\delta = \pi/4$ and clause (i) the Gevrey membership shared with the exponent row.*

Table 1: Per-λ^2 fits of (6) (full tanh form, mode-limited points). The onset constant c clusters near $4/\pi = 1.273$; the exponent q decreases through 1 near $\lambda^2 \approx 13$, matching the archimedean crossover. $\lambda^2 = 200$ (sparse, 8 points) is shown but excluded from the regression.

λ^2	7	13	20	25	50	100	150	200
q	1.185	0.985	0.900	0.880	0.835	0.785	0.760	0.805
c	1.20	1.25	1.28	1.29	1.31	1.28	1.26	1.31

5.3 The prime correction: $q(\lambda^2)$

The leading rate gives $q = 1$. The finite Euler product $\prod_{p\leq\lambda^2}(1-p^{-s})^{-1}$ modulates the archimedean envelope by an oscillatory prime term $S(z) = \sum_{p\leq\lambda^2} p^{-1/2}\cos(z\ln p) + \cdots$ of mean-square strength

$$\langle S^2\rangle = \tfrac{1}{2}\sum_{p\leq\lambda^2} p^{-1} = \tfrac{1}{2}(\ln\ln\lambda^2 + M) + o(1)$$

by Mertens' theorem (M the Mertens constant); equivalently the full log-product is $\sum_{p\leq\lambda^2} -\ln(1-1/p) = \gamma + \ln\ln\lambda^2$. More primes roughen the eigenvector and lower the analyticity exponent below 1. Modeling the reciprocal exponent as a baseline plus the prime fluctuation,

$$\frac{1}{q} = c_0 + c_1\ln\ln\lambda^2, \qquad c_0 = \gamma \text{ (archimedean baseline)}, \quad c_1 = \tfrac{1}{2} \text{ (variance prefactor)}, \tag{11}$$

which gives $q(\lambda^2) = 2/(2\gamma + \ln\ln\lambda^2)$. The numerator 2 is $1/c_1$, the reciprocal of the $\frac{1}{2}$ in the mean square; the $\ln\ln$ is Mertens.

We test (11) against per-λ^2 fits of the full form (6) (Table 1). A linear regression $1/q = c_0 + c_1\ln\ln\lambda^2$ over $\lambda^2 \in \{7, 13, 20, 25, 50, 100, 150\}$ yields

$$c_0 = 0.556\ (\gamma = 0.577), \qquad c_1 = 0.478\text{–}0.508\ (\tfrac{1}{2} = 0.500), \qquad R^2 = 0.98.$$

Euler–Mascheroni γ appears as the intercept of a regression that does not assume it; the c_1 interval spans the regression variants used. The offset of the parameter-free curve is systematic, not centered scatter. Taking $c_0 = \gamma$ exactly, the idealized $q = 2/(2\gamma + \ln\ln\lambda^2)$ runs 5–7% *below* the per-λ^2 fits across the whole range (e.g. 1.10 vs 1.185 at $\lambda^2 = 7$; 0.75 vs 0.785 at $\lambda^2 = 100$), and the regression intercept 0.556 sits 4% under $\gamma = 0.577$, always on the same side. We therefore report both curves side by side — the fitted $(c_0, c_1) = (0.556,\ 0.478\text{–}0.508)$ and the idealized $(\gamma, \frac{1}{2})$ — and the identification of the intercept with γ is a structural hypothesis the data supports to that one-signed accuracy, not a measurement centered on it. *Status: T3; the remaining lemma is a rigorous variance-to-exponent map, now precisely posed on the Gevrey scale (Section 5.1): prove that the almost-periodic prime multiplier S, of mean square $\frac{1}{2}\sum_{p\leq\lambda^2} p^{-1}$, degrades the Gevrey index of the strip-analytic (G^1) archimedean part linearly in that mean square — i.e. $\xi \in G^\mu$ sharply with $\mu = \gamma + \frac{1}{2}\ln\ln\lambda^2$. The characterization of G^μ by coefficient decay* $\exp(-c|k|^{1/\mu})$ *[23] makes this statement equivalent to the measured tail; what is missing is the subordination mechanism, a piece of new analysis rather than a citation.*

5.4 The Shannon coordinate

Lemma 4 (Coordinate from ceiling over slope)**.** *The natural coordinate of the ramp is $N/(\lambda^2\ln\lambda^2)$. The scale at which the leading ramp meets its ceiling is*

$$N^* = \frac{D_{\text{Primes}}}{\partial D/\partial N} = \frac{(4\pi/\ln 10)\lambda^2}{\pi^2/(\ln 10\,\ln\lambda^2)} = \frac{4}{\pi}\,\lambda^2\ln\lambda^2.$$

The λ^2 comes from the prime ceiling (Section 6); the $\ln\lambda^2$ from the inverse of the archimedean slope (10). The combination $\lambda^2\ln\lambda^2$, the time–frequency Shannon number of the box (Landau–Pollak [27]; Slepian–Pollak [26]; Landau–Widom [25]), is therefore forced, not posited. (Measured saturation occurs at $N/(\lambda^2\ln\lambda^2) \approx 2.0$–$2.3$, larger than the linear value $4/\pi$ because the $q < 1$ ramp reaches its ceiling later than the initial-slope line.) The form and the constant separate cleanly: writing $N^* = \lambda^2\ln\lambda^2/\delta$, the grouping $\lambda^2\ln\lambda^2$ is forced by the ceiling's λ^2 scaling and the basis interval $L = \ln\lambda^2$ (Lemma 1) alone, independently of the archimedean constant, while only the multiplier $4/\pi = 1/\delta$ carries the rate δ. That multiplier is the data-pinned piece: the full-ramp onset constant clusters at $4/\pi$ across the cutoffs of Table 1 (mean ≈ 1.27). *Status: the coordinate form is T2 (ceiling-over-slope inputs); the multiplier $4/\pi$ is T3, data-pinned.*

6 The prime ceiling D_{Primes}

The prime channel of the intrinsic law (1) is the accuracy at infinite N and P, set solely by which primes $p \le \lambda^2$ enter the form. It is the relative error of the first zero at the cutoff,

$$\varepsilon_{\text{primes}}(\lambda^2) = \frac{|\nu_1(\infty,\lambda^2) - \gamma_1|}{|\gamma_1|}, \tag{12}$$

with $\nu_1(\infty,\lambda^2)$ the first zero at infinite basis size and fixed λ^2: it is what remains once modes and precision are exhausted, and its digit measurement $D_{\text{Primes}} = -\log_{10}\varepsilon_{\text{primes}}$ is the ceiling the minimum (2) consumes. We fix its leading rate through the plunge *eigenvalue*, the tractable route, kept here in digit space to reinforce the intrinsic statement with the measured quantity.

The plunge eigenvalue gives an *eigenvalue floor* $D_{\text{eig}} = -\log_{10}\varepsilon^\infty(\lambda)$, the digit value of the smallest eigenvalue. The CCM construction ties ε^∞ to the principal prolate spheroidal eigenvalue at bandwidth λ; the prolate eigenvalue asymptotic (Connes [4], §6.4; Fuchs [24]; cf. Connes [3] and Hedenmalm [6]) gives, for the relevant index,

$$1 - \chi(\lambda) \sim \frac{2^{14}\sqrt{2}\,\pi^5}{3}\,\exp\!\big(-4\pi\lambda^2 + \tfrac{9}{2}\ln\lambda^2\big),$$

so that

$$D_{\text{eig}}(\lambda^2) = K\lambda^2 - \frac{9}{2\ln 10}\ln\lambda^2 - C_0, \qquad K = \frac{4\pi}{\ln 10} = 5.45750\ldots, \tag{13}$$

with every constant closed-form: the leading $K = 4\pi/\ln 10$ is the base-10 rendering of the $e^{-4\pi\lambda^2}$ prolate decay rate that appears in Connes' own framework, the $\frac{9}{2}$ subleading-log coefficient is the Fuchs index, and $C_0 = \log_{10}\!\big(2^{14}\sqrt{2}\,\pi^5/3\big) = 6.3736\ldots$ follows from the prolate asymptotic (Groskin [7] independently evaluates the same §6.4 prefactor and obtains $\log_{10} \approx 6.37$, in agreement). This fixes the leading rate of $\varepsilon_{\text{primes}}$: the plunge sets $K\lambda^2$.

One direction of this rate can be anchored rigorously, and we record it as a proposition. Recall that the bottom of the spectrum of the continuum form is a variational infimum, $\varepsilon^\infty(\lambda) = \inf_\varphi QW_\lambda(\varphi,\varphi)/\|\varphi\|^2$, so *any* trial vector bounds it from above, irrespective of the sign of ε^∞.

Proposition 3 (Variational one-sided floor bound)**.** *Let k_λ be the prolate trial vector of the CCM prolate transfer ([1, §§7–8]; [12]), built from the principal prolate spheroidal function at bandwidth λ. Then, unconditionally,*

$$\varepsilon^\infty(\lambda) \;\le\; R[k_\lambda] := \frac{QW_\lambda(k_\lambda,k_\lambda)}{\|k_\lambda\|^2}.$$

Proof. The bottom of the spectrum is the infimum of the Rayleigh quotient over the form domain; any admissible trial vector bounds the infimum from above. □

The proposition is the variational principle verbatim, and is T1 with no caveat. The content is the size of the right-hand side, which we now bound. Take $k_\lambda = \psi_0$, the principal prolate spheroidal mode at time–bandwidth $c = 2\pi\lambda^2$ transferred into the V_n basis. On the localized form's split $QW_\lambda = A_{\mathrm{arch}} - A_{\mathrm{prime}}$ (archimedean part $w_{02} - w_r$, prime part the prime-power sum w_p; [1, §6]) the quotient separates,

$$R[k_\lambda] = \underbrace{\frac{\langle A_{\mathrm{arch}}\psi_0, \psi_0\rangle}{\|\psi_0\|^2}}_{=1-\chi_0(c)} - \underbrace{\frac{\langle A_{\mathrm{prime}}\psi_0, \psi_0\rangle}{\|\psi_0\|^2}}_{=\rho_p(\lambda)}, \tag{14}$$

where the archimedean term equals the prolate *deficiency* $1-\chi_0(c)$ under the band-restricted identification of [1, §6.4], which we take as a hypothesis alongside the sign condition below, and $\rho_p(\lambda)$ is the prime Rayleigh quotient on ψ_0.

Lemma 5 (Prolate floor bound)**.** *If the prime Rayleigh quotient on the principal prolate mode is non-negative, $\rho_p(\lambda) \geq 0$, then*

$$\varepsilon^\infty(\lambda) \;\leq\; 1 - \chi_0(2\pi\lambda^2).$$

Proof. By Prop. 3 and (14), $\varepsilon^\infty(\lambda) \leq R[k_\lambda] = (1-\chi_0(c)) - \rho_p(\lambda) \leq 1 - \chi_0(c)$, the last step by $\rho_p \geq 0$. $\square$

The hypothesis $\rho_p \geq 0$ is a single sign, far weaker than evaluating ρ_p, and it is the natural one: the plunge decomposition $\varepsilon^\infty = (1-\chi_0) - \rho_p$ with $\varepsilon^\infty > 0$ already forces $\rho_p \leq 1 - \chi_0$, and the paper's measured prime contributions on the ground state are positive and $O(1)$ (Section 9). With the Fuchs asymptotic [24] for the principal deficiency,

$$1 - \chi_0(2\pi\lambda^2) = \tfrac{2^{14}}{3}\sqrt{2}\,\pi^5 \exp\!\big(-4\pi\lambda^2 + \tfrac{9}{2}\ln\lambda^2\big)(1+o(1)),$$

Lemma 5 yields the one-sided floor bound

$$D_{\mathrm{eig}}(\lambda^2) \;\geq\; K\lambda^2 - \frac{9}{2\ln 10}\ln\lambda^2 - C_0, \qquad K = \frac{4\pi}{\ln 10}, \quad C_0 = \log_{10}\!\big(\tfrac{2^{14}}{3}\sqrt{2}\,\pi^5\big),$$

whenever $\varepsilon^\infty(\lambda) > 0$. This is now a proof, not a stated expectation: the leading coefficient K and the subleading Fuchs slope $\frac{9}{2}$ are *rigorously bounded from below* on the floor, modulo only the sign hypothesis $\rho_p \geq 0$ (checkable directly through the archimedean/ prime Rayleigh split of [1, §6]) and the classical Fuchs asymptotic. The reverse inequality, that the ceiling cannot be beaten, i.e. a *lower* bound on ε^∞ and on $|\nu_1(\infty, \lambda^2) - \gamma_1|$ of the same exponential order, remains open, and is the archimedean–prime cancellation of Section 9.

Where finite-λ^2 control (rather than asymptotics) is needed, for instance the curvature that produces the empirically steeper matching-floor slope $P_m \approx 5$ below, the modern non-asymptotic prolate literature supplies two-sided eigenvalue bounds with explicit constants: the three-region estimates of Bonami–Jaming–Karoui [15], the improved bounds of Karnik–Romberg–Davenport [16], the upper bounds of Osipov [17], the super-exponential decay estimates of Bonami–Karoui [18], and Kulikov's exponential lower bound before the plunge region [19]. These replace the asymptotic-only inputs wherever a quantified inequality at a specific cutoff is wanted.

The eigenvalue depth is a scaffold, not the object. What the law consumes is the zero error $\varepsilon_{\mathrm{primes}}$, which sits a few digits below the eigenvalue floor: converting the plunge depth ε^∞ into the zero displacement $|\nu_1 - \gamma_1|$, and normalizing by $|\gamma_1|$ (the $\log_{10}|\gamma_1| \approx 1.15$ digits of the relative error), gives the matching floor

$$D_{\mathrm{Primes}}(\lambda^2) = K\lambda^2 - P_m\log_{10}\lambda^2 - c_m, \qquad P_m \approx 5, \quad c_m \approx 9.5. \tag{15}$$

The leading $K\lambda^2$ is shared with the eigenvalue floor, the same plunge rate, so the intrinsic content is a *single* channel, not two competing ceilings; the subleading P_m, c_m are the eigenvalue-to-zero conversion together with the $|\gamma_1|$ normalization. Equivalently the *conversion gap* $\mathrm{conv}(\lambda^2) := D_{\mathrm{eig}} - D_{\mathrm{Primes}} = (P_m - \frac{9}{2})\log_{10}\lambda^2 + (c_m - C_0) \approx \frac{1}{2}\log_{10}\lambda^2 + 3$ is that conversion, read in digit space. The leading rate K is derived (prolate transfer, one side now anchored by Prop. 3); the conversion, hence P_m and c_m, is empirical; of it, the ≈ 1.15-digit part of c_m is the explicit $|\gamma_1|$ normalization and the remainder is the depth-to-zero factor. A structural route to that factor exists, and we record it as a proposal rather than claim it as a derivation. The zero-displacement identity of Lemma 2, applied at the cutoff, gives $|\nu_1(\infty, \lambda^2) - \gamma_1| \le 2\,\eta_\lambda/|\widehat{\xi}'(\gamma_1)|$, with η_λ the plunge-scale perturbation of the transform. The numerator carries the plunge depth through the determinant identity $\det_{\mathrm{reg}}(D - z) = -i\lambda^{-iz}\widehat{\xi}(z)$ of [1, Thm. 5.10]; the denominator carries the L-dependent normalization of the transform and the local zero density $\rho_Q(\gamma_1)$ of Section 7. Deriving the λ-scaling of that ratio would promote (P_m, c_m) from empirical to structural; until that is done the row remains T4, with the route named.

The arithmetic does not move the eigenvalue-floor slope. At the scale the operator resolves, the prime part of the Weil form has large-$|t|$ behavior $S(t) = \frac{1}{2}t^2 - \gamma|t| + c_0$ (Mertens' theorems with prime-number-theorem partial summation). On the even, mean-zero ground eigenvector each piece is inert on the prefactor power: $\frac{1}{2}t^2$ acts as a rank-one *odd* operator that annihilates the even ground state; $-\gamma|t|$ equals $2\gamma K_a$, a multiple of the metric operator $K_a = (-\Delta)^{-1}$, so it only rescales the eigenvalue and shifts the additive constant; and c_0 projects out of the mean-zero subspace (the screw-function operator split of [5]). None of the leading prime terms can therefore supply a $(\lambda^2)^\Delta$ prefactor correction: given the rate, the prefactor power is the Fuchs index $\frac{9}{2}$ of the prolate model (Section 6), and the entire arithmetic contribution is confined to the additive constant. This inertness is a statement about the prefactor only; the archimedean part of the form does not by itself carry the plunge (it is $O(1)$-indefinite; see the rate-origin discussion in Section 9). The empirically steeper matching-floor slope $P_m \approx 5$ is a finite-λ^2 curvature effect, not a change in the asymptotic slope.

Status: the leading coefficient K and the eigenvalue-floor slope $\frac{9}{2}$ ride the prolate transfer (T2), with one side anchored by the variational bound of Prop. 3 (the named lemma being the trace-formula evaluation of $R[k_\lambda]$) and the reverse inequality open (new analysis: the archimedean–prime cancellation); the arithmetic is provably inert on the slope; only the matching-floor offset (P_m, c_m), the eigenvalue-to-zero conversion, is empirical (T4), with a structural route named above.

The floor is smooth, not a prime staircase. A natural hypothesis, that D_{Primes} steps at prime entries, is falsified directly. With N and P held above floor ($N = 300$, $P = 200$), sweeping λ^2 in unit steps from 4 to 30 gives a smooth rise of ≈ 5.3 digits per unit λ^2; the per-unit rate at prime-power entries is statistically indistinguishable from non-prime entries. The ceiling is a smooth function of λ^2, consistent with the analytic form (15).

Floor anchors. Floor-isolated measurements (both N and P above floor) give $D_{\mathrm{Primes}}(100) = 526.08$ at $(N, P) = (1000, 700)$ and a saturated $D_{\mathrm{Primes}}(200) \approx 1070.4$ at $N \ge 2400$. The latter sits at the matching floor $K\cdot 200 - 5\log_{10} 200 - 9.5 = 1070.5$, confirming the two-floor structure at the largest cutoff measured. Regarding provenance, the $\lambda^2 = 200$ anchor was measured against the 2500-digit Arb reference of Definition 1, not a shorter tabulation. A matching-digit count cannot exceed the length of its reference: against a ~ 1000-digit reference the measurement would *saturate* near 999 and could never read 1070.4, so the observed value is itself incompatible with a reference-limited artifact. Such an artifact would present as a suspicious plateau at the reference length, not a reading 70 digits past it.

7 The equilibrium-measure edge

Sections 5 and 6 derived the mode ramp and the prime ceiling by two separate classical routes: the Landau–Widom transition width for the ramp, and the Fuchs prolate tail for the ceiling. This section states the structural fact that ties them together — both are features of a single spectral edge, so the intrinsic law (1) is one edge rather than two coincident mechanisms. It adds no new constant; it is the reason the two budgets share a form.

7.1 The archimedean equilibrium measure

Treat the limiting ordinates $\{\nu_k\}$ as a unit charge in the external field set by the archimedean weight of the Weil form. As $\lambda^2 \to \infty$ the macroscopic distribution is the equilibrium measure μ_Q minimizing the weighted logarithmic energy

$$I[\mu] = \iint \log \frac{1}{|s-t|}\, d\mu(s)\, d\mu(t) + 2 \int Q(t)\, d\mu(t), \qquad \mu \geq 0,\ \textstyle\int d\mu = 1,$$

with external field Q the archimedean potential whose derivative is the phase rate $Q'(t) = \frac{1}{2}\log(t/2\pi)$ of Section 5.2. That a limiting spectral or zero density is the equilibrium measure of such a weighted energy problem is classical (Saff–Totik [28]; in the orthogonal-polynomial and random-matrix setting, Deift [29]); we invoke it here as a known tool. The Euler–Lagrange condition $2\int \log|s-t|\, d\mu(t) = 2Q(s) + \text{const}$ on $\operatorname{supp}\mu$ fixes the density; with this field it is

$$\rho_Q(t) = \frac{1}{2\pi} \log \frac{t}{2\pi}, \tag{16}$$

the Riemann–von Mangoldt zero-counting density ($\int_0^T \rho_Q = \frac{T}{2\pi}\log\frac{T}{2\pi e}$). The smooth backbone of the spectrum is thus the equilibrium measure of the archimedean field, recovered without reference to the primes and independent of RH.

The variational identification can be re-based on Szegő theory, which turns the open step into a *named* lemma. The archimedean part of the truncated matrix is a Toeplitz-type operator whose symbol is explicit, the phase rate $\partial_t\theta(t) = \frac{1}{2}\log(t/2\pi) + O(t^{-2})$ recorded in [1, eq. (3.24)], and first-order Szegő / Kac–Murdock–Szegő asymptotics for operators of this symbol class [21, 22] yield the bulk eigenvalue-counting law directly, which is precisely the density (16). The arithmetic part is a *bounded* perturbation uniformly in N (the content of the discreteness argument in [1, Thm. 3.6]: the contribution of the non-archimedean places to the operator is bounded), so Weyl's inequality transfers the bulk law to the full form unchanged up to an $O(1)$ shift. This two-step route — first-order Szegő for the log symbol, plus bounded perturbation — replaces the variational analogy with checkable classical inputs; the second-order term of the same theory [25, 22, 20] is then the transition width of Section 7.2. *Status: T2 — the density* (16) *is the standard archimedean main term, the bounded-perturbation step is proven in [1], and the remaining open step is the named first-order Szegő lemma for this symbol class applied to the truncated operator; the tier carried by the Szegő route above.*

7.2 One edge, two scales

The equilibrium measure has a lower band edge: the eigenvalues descend from the $O(1)$ bulk into the deep, positive plunge (positivity forced by the Weil form). That edge is a single object, and the two budgets are its two scales:

- *the transition is the mode ramp* — the width of the crossover from bulk to plunge, as the resolved dimension crosses the Shannon number $\lambda^2 \ln \lambda^2$, is the Landau–Widom log-width [25], which is the tanh ramp of D_{nModes} (Section 5);

- *the extreme tail is the prime ceiling* — the depth past the edge is the Fuchs extreme-value asymptotic [24], which is D_{Primes} (Section 6).

So the Landau–Widom width and the Fuchs tail, invoked separately above, are the transition and the tail of the *same* band edge. This is the mechanism behind the algebra $D_{\text{nModes}} = D_{\text{Primes}} \cdot g$ of Section 3.1: D_{nModes} and D_{Primes} are one edge D_{edge} read at two resolutions, and the intrinsic error (1) is correspondingly one edge: its finite-N transition $\varepsilon_{\text{modes}}$ and its extreme tail $\varepsilon_{\text{primes}}$ are the ramp and the ceiling of μ_Q's lower edge. The unification is conceptual, not a new estimate: the microscopic universality class of the edge (Airy, Bessel, or a Landau–Widom band edge) and the exact deep-edge constant remain open, the latter being the same archimedean–prime cancellation left open for D_{Primes} in Section 6. The equilibrium-measure framework itself is classical (above); to the best of the author's knowledge, its application to the Connes–Consani–Moscovici / Connes–van Suijlekom Weil form (identifying the zero-counting backbone as the archimedean equilibrium measure, and the mode ramp and prime ceiling as the transition and extreme tail of that measure's lower edge) is new here. *Status: T3 — structural unification riding the prolate transfer already used for both budgets; the bulk beneath it is now T2 via the Szegő route of Section 7.1, while the edge universality class and the deep-edge constant remain open (new analysis).*

8 The complete formula

8.1 The assembled law

Collecting Sections 4–7, the law stands in the two forms of the introduction. *Intrinsically* it is the single edge of Section 7: the relative error obeys the two-channel decomposition

$$E(N, \lambda^2) \le \varepsilon_{\text{modes}}(N, \lambda^2) + \varepsilon_{\text{primes}}(\lambda^2) \tag{17}$$

(an equality up to sign cancellation; Prop. 1); equivalently, in digit units, the edge function $D_{\text{edge}}(N, \lambda^2) = D_{\text{Primes}}(\lambda^2) \cdot \tanh\big((c\,N/(\lambda^2 \ln \lambda^2))^q\big)$, whose $N \to \infty$ ceiling $D_{\text{edge}}(\infty, \lambda^2) = D_{\text{Primes}}$ is the prime floor and whose finite-N transition is the mode ramp D_{nModes}. *Observed* at working precision P, it is that one edge capped by the instrument resolution:

$$D(\lambda^2, N, P) = \min\Big(\underbrace{P + O(1)}_{D_{\text{Wprec}}},\ \underbrace{D_{\text{Primes}} \tanh\big((c\,N/(\lambda^2 \ln \lambda^2))^q\big)}_{D_{\text{nModes}} = D_{\text{edge}}},\ \underbrace{K\lambda^2 - P_m \log_{10} \lambda^2 - c_m}_{D_{\text{Primes}} = D_{\text{edge}}(\infty)}\Big), \tag{18}$$

with $K = 4\pi/\ln 10$ exact, $q = 2/(2\gamma + \ln \ln \lambda^2)$, and $c \approx 4/\pi$. The last two entries are the one edge D_{edge} at finite N and at $N \to \infty$, so the operative law is $D = \min(D_{\text{Wprec}}, D_{\text{edge}}) + O(1)$: a single spectral edge, capped by precision. This is the Andrews CCM Convergence Law. Of its parts, the minimum bound, the precision slope, the stretched-exponential rate unification (Prop. 2), and the zero-displacement transfer (Lemma 2) are rigorous; every remaining component carries the explicit tier, open item, and open-item type recorded in Table 2, from the T2 constants through the T3 exponent and onset constant to the single empirical offset (T4).

To the best of the author's knowledge, the complete convergence model of (18) is new: the assembly of the three budgets into a single law, the closed-form mode exponent $q(\lambda^2) = 2/(2\gamma + \ln \ln \lambda^2)$, the Shannon coordinate $\lambda^2 \ln \lambda^2$, the stretched-exponential identification of the mode budget, and the recognition of the mode ramp and prime ceiling as one equilibrium-measure edge appear here for the first time. The rate it characterizes is precisely the problem named open in [1]: Groskin [7] measures a Galerkin exponent but disclaims deriving it, and Śliwiński [8] establishes only a lower bound, so no prior work supplies the functional form given here. The novelty claimed is the law as a whole and the identification of its mechanism; the underlying construction [1, 2], the Paley–Wiener strip mechanism [7], and the classical prolate and Mertens inputs are credited to their sources throughout.

8.2 Convergence criterion

Proposition 4 (Convergence criterion and the prime bottleneck)**.** *Let $\nu_1 = \nu_1(\lambda^2, N, P)$ and $D = -\log_{10}(|\nu_1 - \gamma_1|/|\gamma_1|)$. Assume, at each fixed cutoff considered, the finite-cutoff hypothesis*

$$(\mathrm{H}_\lambda): \qquad \varepsilon^\infty(\lambda) > 0 \quad \textit{and} \quad \nu_1(\infty, \lambda^2) \neq \gamma_1. \tag{19}$$

(i) (Prime bottleneck.) *For every fixed λ^2 satisfying* (H_λ), $\sup_{N,P} D(\lambda^2, N, P) = D_{\mathrm{Primes}}(\lambda^2) < \infty$*. Hence no sequence with bounded λ^2 has $\nu_1 \to \gamma_1$: convergence to the zero forces $\lambda^2 \to \infty$.*

(ii) (Criterion.) *Since $D_{\mathrm{nModes}} \le D_{\mathrm{Primes}}$ (a mode-limited count cannot beat the ceiling) $D \to \infty$ if and only if $D_{\mathrm{Wprec}} \to \infty$ and $D_{\mathrm{nModes}} \to \infty$; the latter forces $\lambda^2 \to \infty$ (part (i)), with N past the Shannon scale $\lambda^2 \ln \lambda^2$ and $P \to \infty$. The mode budget rises monotonically toward its ceiling as $N \to \infty$ [1, Prop. 3.4].*

Consequently, conditional on the continuum plunge $D_{\mathrm{Primes}}(\lambda^2) \to \infty$ (i.e. $\varepsilon^\infty(\lambda) \to 0$), convergence holds along any path $\lambda^2 \to \infty$ with $N \gtrsim (4/\pi)\lambda^2 \ln \lambda^2$ and $P \gtrsim K\lambda^2$, at rate $D \sim K\lambda^2$.

Proof. (i) As $N \to \infty$ the truncation converges to the continuum operator at cutoff λ, whose first zero lies at relative distance $10^{-D_{\mathrm{Primes}}(\lambda^2)}$ from γ_1; here $D_{\mathrm{Primes}}(\lambda^2) < \infty$ by hypothesis (H_λ) and is N-independent [5, Thm. 1.3], and finite working precision only lowers D. Hence $D \le D_{\mathrm{Primes}}(\lambda^2)$, finite, for all N, P. (ii) By Prop. 1, $D = \min(D_{\mathrm{Wprec}}, D_{\mathrm{nModes}}, D_{\mathrm{Primes}}) + O(1)$; since $D_{\mathrm{nModes}} \le D_{\mathrm{Primes}}$, this is $\min(D_{\mathrm{Wprec}}, D_{\mathrm{nModes}}) + O(1)$, so $D \to \infty$ if and only if $D_{\mathrm{Wprec}} \to \infty$ and $D_{\mathrm{nModes}} \to \infty$; since $D = -\log_{10}(|\nu_1 - \gamma_1|/|\gamma_1|)$, this is exactly $\nu_1 \to \gamma_1$. □

Status of the hypothesis (H_λ)**.** We state (H_λ) explicitly because finite-cutoff Weil positivity is *not* known unconditionally for all λ: [1] is explicit that positivity of the continuum measure cannot be asserted in general, and positivity for all λ is precisely the open conjecture linking the construction to RH. What is unconditional is the small-support range: Yoshida proved the Weil hermitian form positive definite for test functions of sufficiently small support [9], Bombieri reproved and quantified this in his study of the truncated functional [10], Connes–Consani gave the operator-theoretic strengthening at the archimedean place via the Sonin space [11], and the ζ-cycles paper [12] tracks finite-λ positivity prime by prime through the first cutoffs. Beyond that range (H_λ) is verified numerically at every cutoff measured in this program ($\lambda^2 \le 200$, and $\lambda^2 \le 1000$ in [31]); the second clause $\nu_1(\infty, \lambda^2) \neq \gamma_1$, that the truncated Euler product does not accidentally reproduce the zero exactly, is likewise numerical, and is what the finiteness of D_{Primes} actually uses.
This refines the qualitative monotone convergence of [1, Prop. 3.4] into an explicit, conditional rate. The equivalence $D \to \infty \Leftrightarrow \nu_1 \to \gamma_1$ is definitional and rigorous (T1); the prime bottleneck (i) is rigorous *conditional on* (H_λ), which is unconditional in the Yoshida–Bombieri small-support range and numerical beyond it; the conditional rate inherits the rigor of D_{Primes}: the prolate transfer (T2, one side anchored by Prop. 3) and the continuum plunge it assumes.

Conceptually, the criterion isolates the prime cutoff as the sole hard bottleneck: precision and basis size are freely improvable and impose no ceiling, so reaching the exact zero is, at root, a statement about $\lambda^2 \to \infty$ and the growth of the prime content. The rate $D \sim K\lambda^2$ quantifies the cost — each additional digit demands a fixed increment $\ln 10/(4\pi)$ in λ^2 — turning the construction's qualitative convergence into a concrete, if conditional, budget.

9 Discussion

Relation to Groskin [7]. Groskin independently implemented the Connes–van Suijlekom matrix and measured a Galerkin convergence exponent $s(c)$ growing with the cutoff, proposing a Paley–Wiener analyticity-strip mechanism. The present work builds on that mechanism: Prop. 2(ii) shows his $s(c)$ is the local slope of our stretched-exponential, and the strip width is identified with the archimedean $\Gamma_{\mathbb{R}}$ rate $\delta = \pi/4$. What is new here is the digit-space formulation, the Shannon coordinate, the closed form for $q(\lambda^2)$, and the embedding of the rate in the full three-budget minimum.

Groskin's revised analysis, extended to cutoff $\lambda^2 = 100$, is independent corroboration of this two-regime structure. He finds that no single smooth-convergence law fits his eigenvalue-space data and explicitly declines to assert a convergence rate, reporting instead an unresolved tension between a *drifting* effective exponent and an apparent *finite* convergence limit. Both halves are accounted for here: Prop. 2(ii) explains the drift (a fixed Sobolev index cannot persist, since the local exponent $2s_{\mathrm{loc}}(N) = aqN^q$ grows with N), while the prime ceiling of Section 6 supplies the finite limit $D_{\mathrm{Primes}}(\lambda^2)$. What looks like an irreconcilable choice within a single-rate framework is simply the pre-saturation ramp and the ceiling of the minimum law (2), seen in one dataset.

Relation to Śliwiński [8] and Suzuki [5]. Śliwiński proves an inverse-logarithmic lower bound on the *mean* spectral error over the whole spectrum (Thm. 3.1), a different regime from the first-zero convergence here, and consistent with it, since the mean is dominated by the unconverged high-index eigenvalues while $\nu_1 \to \gamma_1$ converges super-exponentially. Suzuki's screw-function analysis supplies the archimedean $\ln|z|$ multiplier and the prime-frequency support that underlie Sections 5.2–5.3.

Origin of the leading rate: arithmetic, not archimedean. The prime ceiling is *modeled* by an archimedean object, the prolate eigenvalue of Section 6, yet the exponential rate is not carried by the archimedean part of the Weil form itself. Diagonalizing the screw-function split [5] at high precision, the archimedean-only localized form is $O(1)$-indefinite, with no exponentially small eigenvalue. Restricting it to the Sonin subspace, the orthogonal complement of the band-concentrated prolate modes, where Connes–Consani establish archimedean Weil positivity [11] (surveyed in [4]), restores positivity, but only as an $O(1)$ gap: the smallest Sonin eigenvalue is 2.26, 2.28, 2.57 at $\lambda^2 = 5, 8, 13$, essentially flat in λ^2, against a floor $K\lambda^2 = 27, 44, 71$. A direct diagonalization of the archimedean-only form at $\lambda^2 = 13$–50 finds the same structure from the eigenvector side: the lowest even mode is a delocalized Sonin-type state with an $O(1)$ eigenvalue, so the band-concentrated plunge eigenvector exists only for the full form. Archimedean positivity and the plunge are therefore *decoupled*: the place at infinity supplies an $O(1)$ spectral gap, while the exponentially small $\varepsilon^\infty \sim e^{-4\pi\lambda^2}$ is produced by the cancellation between the archimedean and prime parts of the full form. This locates the single open analytic step behind the leading coefficient: a quantitative archimedean–prime cancellation estimate on the even ground eigenvector, the precise sense in which K is T2.

Practical use. The law is a parameter-selection guide: to target D^* digits, pick $\lambda^2 \gtrsim D^* \ln 10/(4\pi) = D^*/5.46$ so the ceiling clears D^*; pick N so the tanh ramp clears D^* (a few times the Shannon number $\lambda^2 \ln \lambda^2$); and pick $P \gtrsim D^*$. Increasing any one parameter past the binding minimum yields nothing.

10 Rigor ledger

Table 2 collects the rigor status of every component of the law, together with the open item that would promote each to the next level. A fourth column types each open item: *named lemma* means the missing step is a precisely posed statement inside an existing classical theory (variational principle, Rouché, Szegő, Gevrey characterization), so that closing it is a matter of executing known machinery; *new analysis* means genuinely new estimates are required. Four rows sit in the second category.

Table 2: Rigor status of each component. T1: rigorous. T2: derived modulo one named lemma. T3: derived in structure and validated against data, with a map not yet rigorous. T4: empirical. Open-item type: *named lemma* = precisely posed inside existing theory; *new analysis* = new estimates required.

Component	Status	Open item	Type
Minimum structure (Prop. 1), lower bound	**T1**	—	—
Minimum structure, matching upper bound	**T1***	cancellation excluded at seams	empirical check
$D_{\mathrm{Wprec}} = P + O(1)$, slope 1	**T1**	implementation offset	—
Stretched-exp rate unification (Prop. 2)	**T1**	—	—
Zero-displacement transfer (Lem. 2)	**T1**	—	—
Tail-to-transform transfer (Lem. 3, Cor. 1)	**T1**	—	—
Convergence criterion (Prop. 4)	**T1** | (H_λ)	positivity beyond Yoshida range	new analysis
D_{Primes} leading K: floor lower bound	**T2**	prime sign $\rho_p \geq 0$ (Lem. 5)	named lemma
D_{Primes} leading K: ceiling unbeatable	open	archimedean–prime cancellation	new analysis
D_{Primes} floor slope 9/2	**T2**	Fuchs index (Lem. 5); primes provably inert	named lemma
D_{Primes} matching offset (P_m, c_m)	**T4**	λ-scaling of $\eta_\lambda/\lvert\widehat{\xi}'(\gamma_1)\rvert$, route named	named lemma
Shannon coordinate $\lambda^2 \ln \lambda^2$	**T2**	ceiling/slope (Lem. 4)	named lemma
Equilibrium-measure bulk (RvM density)	**T2**	first-order Szegő for the log symbol	named lemma
Edge unification (one edge, two scales)	**T3**	edge universality class; deep-edge constant	new analysis
Archimedean rate $\delta = \pi/4$	**T2**	Conj. 1(ii) (Lem. 2 closes the transfer)	new analysis
Exponent $q = 2/(2\gamma + \ln\ln\lambda^2)$	**T3**	Gevrey subordination (Conj. 1(i))	new analysis
Onset constant c ($\sim 4/\pi$)	**T3**	exact rational	—

*Unconditional as a lower bound; the equality direction holds absent cancellation between the signed channels (Prop. 1), verified across the experimental program. T1 | (H_λ): rigorous conditional on the finite-cutoff hypothesis (19), unconditional in the Yoshida–Bombieri small-support range [9, 10].

11 Conclusion

We have given a single law (2) for the convergence of the CCM zeta spectral triple. In it the minimum structure is forced; the prime ceiling has the parameter-free leading coefficient $4\pi/\ln 10$, rigorously anchored from one side by the variational bound of Prop. 3 and open in the reverse inequality; and the mode budget, the convergence rate named open in [1], is the logarithm of a stretched-exponential whose exponent is the eigenvector's analyticity class, derived end to end as $q = 2/(2\gamma + \ln\ln\lambda^2)$. The mode ramp and the prime ceiling are not two mechanisms but one object, the lower edge of the archimedean equilibrium measure whose bulk density is the Riemann–von Mangoldt law, seen at two scales (Section 7). Read as a convergence statement, the law says $\nu_1 \to \gamma_1$ exactly when all three budgets diverge, giving an explicit rate $D \sim (4\pi/\ln 10)\lambda^2$ along $\lambda^2 \to \infty$ (Prop. 4). Each claim is stated with an explicit rigor level. The rigorous core (the minimum bound, the precision slope, the stretched-exponential convergence proposition 2 (which unifies the digit- and eigenvalue-space rates), and the zero-displacement and tail-to-transform lemmas 2 and 3) stands on its own; the variational bound of Prop. 3 anchors one side of the prime floor; and the remaining open items are typed in Table 2. Of these, the *named lemmas* (the prime-sign hypothesis $\rho_p \geq 0$ behind the floor bound of Lem. 5 and the first-order Szegő asymptotics for the log symbol) are precisely posed inside classical theories and are the natural next steps; the *new-analysis* items (the coefficient-decay estimate for the $\Gamma_{\mathbb{R}}$-weighted form (Conjecture 1), the Gevrey subordination behind q, the archimedean–prime cancellation behind the unbeatable ceiling, and positivity beyond the Yoshida range) are where the construction's genuinely open mathematics lives.

12 Explicit non-claims

The per-component rigor of the law is the subject of Table 2 and is not repeated here. Beyond it, we are explicit about three boundaries.

- **No proof of the Riemann Hypothesis.** We characterize how the CCM/CvS approximation converges; the link from the construction to RH, continuum positivity of QW_λ for all λ, is the open conjecture of [1, 2, 4], and is neither assumed nor advanced here.
- **We do not establish that the construction reproduces the zeta zeros.** That the finite operator's zeros converge to the true ordinates as $\lambda^2 \to \infty$ is the construction's own open premise; we quantify the approach to γ_1 *under* that premise (Prop. 4) and do not prove the premise itself.
- **Scope.** The law is stated and tested for the first zero γ_1 of the Riemann zeta function. We make no claim about its constants for higher zeros, nor about extension to Dirichlet L-functions or other settings. The rate-law constants (q, c) are pinned on $\lambda^2 \leq 150$ ($\lambda^2 = 200$, sparse at 8 points, is shown but excluded from the q regression); their constancy at large λ^2, the regime in which the law becomes a convergence statement, is untested, and we make no claim that the fitted forms persist beyond the measured range.
- **No unconditional finite-cutoff positivity.** Statements requiring $\varepsilon^\infty(\lambda) > 0$ or $\nu_1(\infty, \lambda^2) \neq \gamma_1$ are conditioned on the hypothesis (H_λ) of eq. (19), which is unconditional only in the small-support range of Yoshida [9] and Bombieri [10] and is verified numerically beyond it. We do not claim positivity of QW_λ for general λ.

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Statement on the use of AI tools

In keeping with the arXiv policy on authors' use of generative AI language tools, the author discloses that Anthropic's Claude AI model served as an aid in preparing the exposition and the Rust implementation code, to specifications set by the author. AI assistance was used to improve prose formatting and readability. It was also used to gather research material, propose candidate arguments, and validate calculations; all such output was reviewed by the author. The author originated, directed, and conducted every computational experiment presented here. The mathematical development was led by the author with assistance from AI. The author bears full responsibility for the contents of this manuscript.

Empirical testing and code availability

The results above rest on an extensive empirical program: over 5,000 high-precision CCM experiments across the cutoff, basis-size, and working-precision axes, very likely the largest set of CCM calculations assembled to date, especially at large configurations. The testing came first and the analysis followed. Candidate rate laws were tested against the data and discarded when they failed [32], and each failure narrowed the space of forms the operator would tolerate. Isolating one parameter at a time for study, with the other two driven out of the way, is how each budget's form was read, and the breadth of the dataset ($\lambda^2 = 200$ at $N \geq 2400$ here, $\lambda^2 = 1000$ in the reproduction [31]) is what made that isolation possible. The extensive empirical testing was not decoration but the heart of how the author mapped the operator's behavior.

The paper source and reproduction scripts are available at `https://github.com/TeamXcelerator/ccm-convergence-rate`; the shared high-precision library (Xcelerator Toolkit) is at `https://github.com/TeamXcelerator/xcelerator-toolkit`. Both are source-available in accordance with the licensing provided with each repository.

A Self-contained structural facts

This appendix records the two structural facts about $\widehat{\xi}$ used in the body, the closed form of the transform (Lemma 1) and entirety/reality of its zeros, in a form that depends only on elementary computation and on the self-contained proofs of [1]. Its purpose is to make explicit that no statement in this paper routes through the Carathéodory–Fejér machinery of [2], which we cite as the historical origin of the reality mechanism only.

A.1 The closed form of the transform, from scratch

Write $t = \ln u \in [-L/2, L/2]$, so that $V_n(u) = L^{-1/2} e^{2\pi i n(t+L/2)/L} = L^{-1/2}(-1)^n e^{2\pi i n t/L}$ and the Fourier–Mellin transform is the Fourier integral over the interval. For a single mode,

$$\widehat{V}_n(z) = \int_{-L/2}^{L/2} e^{izt}\, V_n(t)\, dt = L^{-1/2}(-1)^n \int_{-L/2}^{L/2} e^{i(z+2\pi n/L)t}\, dt$$
$$= L^{-1/2}(-1)^n \, \frac{2\sin\big((z+2\pi n/L)\, L/2\big)}{z+2\pi n/L}.$$

Since $\sin(zL/2 + \pi n) = (-1)^n \sin(zL/2)$, the sign cancels:

$$\widehat{V}_n(z) = \frac{2L^{-1/2}\sin(zL/2)}{z+2\pi n/L}.$$

Summing over the eigenvector $\xi = \sum_{|n|\leq N} \xi_n V_n$ and reindexing $j = -n$ (using $\xi_{-j} = \xi_j$ for the even eigenvector, or simply relabeling) gives

$$\widehat{\xi}_N(z) = 2L^{-1/2}\sin(zL/2) \sum_{|j|\leq N} \frac{\xi_j}{z - 2\pi j/L},$$

the closed form of Lemma 1, in agreement with [1, Prop. 5.9]. The derivation is a one-line Fourier integral; no external input is used. Entirety is immediate: each apparent pole $z = 2\pi j/L$ is cancelled by the zero of $\sin(zL/2)$ there (the removable-pole computation in the proof of Lemma 1), so $\widehat{\xi}_N$ is entire of exponential type $L/2$.

A.2 Reality of the zeros: provenance

The statement that, in the critical case, all zeros of $\widehat{\xi}_N$ are real and coincide with the spectrum of the perturbed operator is [1, Thm. 5.10]. We record why its proof is self-contained there. The proof chain consists of: (a) [1, Lem. 5.2], the observation that the relevant matrices have the structure $\tau_{ij} = (b_i - b_j)/(i-j)$; (b) [1, Lem. 5.4], the elementary commutator identity $DT - TD = |\beta\rangle\langle\eta| - |\eta\rangle\langle\beta|$ for such matrices, with D the diagonal mode operator; and (c) the conclusion of [1, Thm. 5.10]: when ε_N is the smallest eigenvalue with even simple eigenvector, $T = QW_\lambda^N -$

$\varepsilon_N \geq 0$, the perturbed operator is self-adjoint in the T-inner product, and the transform is proportional to a regularized characteristic determinant, $\det_{\mathrm{reg}}(D'' - z) = -i\lambda^{-iz}\widehat{\xi}(z)$, whose zeros are therefore real, being the characteristic polynomial of a self-adjoint matrix. Each of (a)–(c) is proven within [1] by direct computation; the general Carathéodory–Fejér extension of [2], of which this is the specialization, is not invoked at any step used here.

A.3 The independent classical foundation

For completeness we record the pre-CvS lineage on which the truncated Weil form itself rests: Weil's positivity criterion; Yoshida's unconditional positivity for test functions of sufficiently small support [9]; Bombieri's study of the quadratic functional restricted to a finite interval, its minimizer in the L^2 unit ball, and the eigenvalues of its finite truncations [10]; the operator-theoretic strengthening of Yoshida's theorem at the archimedean place by Connes–Consani [11]; the prolate identification of the small eigenvectors in the ζ-cycles paper [12]; and Suzuki's screw-function framework [13, 14, 5], which unifies Yoshida, Bombieri, and the Connes–Consani(–Moscovici) results by means of continuous functions and de Branges-space methods. The construction studied in this paper can be founded entirely on this lineage together with [1].

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